EWB COURSES

AI-POWERED CYBERSECURITY

ASSESSMENT 01

SECURITY FOUNDATIONS

SUBMITTED BY

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Task A | Secure a student account - 25 marks

A1. Identify five distinct risks. For each, name the asset at risk and explain how confidentiality, integrity or availability could be affected. (10 marks)

Risk	Asset	CIA impact	Control	limitation
Short password	Email and College website access	Confidentiality: Both accounts may be accessed.	Use strong passwords.	Phishing can still steal passwords
Shared signed-in account	Files and college website access	Confidentiality and integrity: Others can see or change data.	Can use separate accounts.	May steal able to access if she forget to logout
Unknown login approval	Student account	Confidentiality: Unwanted access may be allowed.	Use MFA only when signing in.	A user may approve by mistake.
Project copy	Project files	Availability: Work may be lost.	Keep secure, regular backups.	Backups must be maintained.
Delayed updates	Laptop and files	Integrity: Malware could alter files.	Install security updates	Updates do not stop every threat.

A2. Give one practical control for each risk. Explain why it helps, including at least one limitation. Prioritise your first three actions and justify the order. (10 marks)

	Risk	**Practical Control**	**Why It Helps**	**Limitation**
1	Using the short password	Mira should create a strong and different password for each account.	If someone finds out one password, they cannot easily access her other account.	Someone could still steal her password through a fake email or website.
2	Approving unknown login requests	She should only approve login requests when she is trying to sign in herself.	This can stop someone else from entering her account without permission.	She may approve a request by mistake if she does not check carefully.
3	Not updating the laptop	She should install updates whenever they are available.	Updates help fix security problems and keep her laptop safer.	Some security problems may still remain even after updating.

4	Keeping projects on one laptop	She should keep another copy of her projects in secure cloud storage or on an external drive.	If her laptop stops working or gets lost, she can still get her project files back.	She might lose her latest work if she forgets to update the backup.
5	Sharing signed-in laptop	She should log out of her account and let her friend use a separate account.	This helps keep her personal files and college information private.	Her friend may still access her account if she forgets to log out.

First Three Priority Actions

1. Change Passwords:Mira should first change her passwords because she is using the same short password for two accounts. If someone guesses it, both accounts could be in danger. Using different passwords will make her accounts safer.
2. Be Careful with Login Requests:Her second step should be to stop approving unknown login requests. She should check every request before approving it. This is important because someone else might be trying to enter her account.
3. Keep Backup Copies:Her third step should be to save another copy of her projects. Right now, she has everything on one laptop. If something happens to it, she could lose her work. Having a backup will help her recover her files.

A3. Explain authentication versus authorisation using Mira's learning portal as your example. (5 marks)

No.	Authentication	Authorisation
1	Authentication means checking if the person logging in is the real user.	Authorisation means deciding what a person can access after logging in.
2	It takes place when Mira tries to log in to her college website.	It takes place after Mira logs in to her account.
3	Mira uses her username and password to log in.	Mira can only open the pages she is allowed to use.

4	It helps protect her account from unknown people.	It stops users from opening information they are not allowed to see.
5	Sometimes Mira needs to enter an OTP for extra security.	Mira can check her notes, assignments and other study materials.
6	If Mira enters the wrong password, she cannot log in.	Even after logging in, Mira cannot change another student's marks.
7	It is the first step in checking a user's identity.	It decides what the user can do after logging in.

Task B: Analyse a suspicious message

B1. Identify five specific warning signs from the evidence. Quote only the relevant phrase and explain why it needs scrutiny. (10 marks)

Warning sign	Explanation
1. "FINAL WARNING - scholarship expires in 20 minutes"	The message is trying to scare the student by giving a very short deadline. It creates pressure and does not give enough time to check whether the scholarship information is true.
2. "Reply-To: help@personal-awards.example"	The reply email address is different from the sender's email address. This makes me suspicious because it does not look like an official college email address.
3. "Enter your college password and OTP"	Asking for a college password and OTP is unsafe. No one should share these details through an unknown link because someone could use them to access the account.
4. "Do not contact the college office"	This statement is suspicious because it tells students not to contact the college for confirmation. A genuine scholarship message should not stop students from checking the information.

5. "Reply with a photo of your identity card"	Asking for an identity card photo can put the student's personal information at risk. Such documents should only be shared through a trusted and official college channel.

B2. Describe a safe verification process using an independently obtained official contact or portal. Explain why HTTPS or a familiar display name alone would not prove the message is genuine. (5 marks)

If I get this kind of message, I will first check whether it is really sent by my college. I will not click on the link or share my password and OTP without confirming it.First, I will visit my college's official website by typing the website address myself. Then I will log in to the student portal and check if there is any scholarship update. If I don't find anything, I will contact the college office using the official phone number or email given on the college website. I will also inform the college if I find the message suspicious.

Why HTTPS or a familiar name is not enough:

Just because a website has HTTPS does not mean it is genuine. It only means that the connection is secure. Also, anyone can use a familiar name like "Scholarship Desk" to make a fake email look real. So, I should always confirm such messages through the official college website or office before trusting them.

B3. Write a short response plan for a student who already entered a password and OTP. Include immediate containment, official reporting and follow-up checks. Do not ask the student to send you the password or OTP. (7 marks)

If a student has already entered their password and OTP on a suspicious website, they should not panic. They should take a few steps quickly to protect their account.

1. Change the password: First, the student should go to the official college website and change their password immediately. If they use the same password on other websites, they should change those passwords too.``
2. Log out from other devices: They should sign out of all devices where their account is currently logged in so that nobody else can access it.
3. Check account security: They should check their login activity and make sure that no unknown device or phone number has been added to their account.
4. Contact the college: They should inform the college IT department about what happened so that the college can help protect their account.
5. Report the fake message: They should report the suspicious email to the college and avoid replying to it again.
6. Check for unusual activity: They should check their emails and student portal for unknown logins, password changes or any other activity they did not perform.
7 Keep monitoring the account: For the next few days, they should regularly check their account to make sure that no further suspicious activity is happening.

B4. State one important fact you cannot establish from this message alone, and what additional evidence would help. (3 marks)

After reading the message, I cannot say for sure whether the email was actually sent by the college or by someone else. The message looks suspicious, but just by reading it, we cannot know who is behind it.To confirm this, I would check with the college office and ask whether they have really sent this scholarship message. I would also check the official scholarship notice and sender's email details.

Task C: Triage a sign-in alert

C1. Build a four-point timeline connecting the logs and user statement. Distinguish observed events from your interpretation. (8 marks)

Time	Observed Event (What happened)	My Interpretation
09:00–09:01	Student07 logged in successfully using the usual IP address (192.0.2.10), and the MFA request was approved.	This seems like a normal login because the student used the usual classroom network.
10:14–10:15	Two login attempts were made from a different IP address (198.51.100.23), but both failed.	It looks like someone was trying to access the account but entered the wrong password.
10:16–10:18	The password was accepted. The first MFA request was denied, but the next request was approved.	This is suspicious because the student said they were not trying to log in but approved the request thinking it would stop the repeated notifications
10.20	A new session was created from the different IP address (198.51.100.23).	This makes me think that someone may have accessed the student's account after the MFA request was approved.

Give a likely explanation and one alternative explanation. List two additional checks that could distinguish them. Avoid claiming the source IP identifies a person. (7 marks)

1. Likely Explanation:

I think someone may have tried to access student07's account using a stolen or guessed password. After two failed attempts, the password was entered correctly. The first MFA request

was denied, but the student later approved another request by mistake. Because of this, the person may have gained access to the account and created a new session.
2. Alternative Explanation:
Another possibility is that the student might have tried to log in using a different device or network but did not realise it. They may have forgotten about the login attempt or accidentally approved the MFA notification.
3. Two Additional Checks:
1. The IT team should check the login history, device details and browser information to find out whether the login came from the student's usual device or an unknown device.
2. The IT team should ask the student whether they were using any other device or network around that time and confirm if they recognise the login activity.

C3. Recommend three prioritised response actions for the authorised IT team. Include preservation of evidence and protection of the account. (6 marks)
The IT team should take the following steps to protect student07's account and find out what happened.
1. Secure the student's account: First, the IT team should immediately block the suspicious login session and ask the student to change their password. They should also check and reset the MFA settings if needed to prevent further unauthorised access.
2. Save all the evidence: The IT team should keep the login records, IP addresses, timestamps and MFA details safely. This information will help them understand what happened during the investigation.
3. Check for further suspicious activity: The IT team should review the student's recent login history and check whether any unknown devices have accessed the account or any account details have been changed

C4. Write a 60-100 word incident summary: evidence, uncertainty, priority and next action. (4 marks)
On 18 October 2026, student07's account showed some unusual login activity. Two password attempts failed, but the third attempt was successful from a different IP address. The MFA request was first denied and later approved by the student by mistake. After that, a new session was created. It is possible that someone accessed the account without permission, but this is not confirmed yet. The IT team should secure the account, save all login records and investigate the suspicious activity.

Task D: Review an AI security answer

D1. Identify and correct five unsafe or unsupported claims/actions in this output. Explain each correction using the evidence or a trustworthy reference. (10 marks)

	Incorrect statement from AI	Correction and Explanation
1.	"The email is safe because it uses HTTPS."	HTTPS does not mean a website is genuine. It only protects the connection between the browser and website. We should check the website through the official college portal before trusting it.
2.	"The person using 198.51.100.23 stole the password."	We cannot say who was using that IP address or how they got the password. The logs only show failed attempts followed by a successful password login. More investigation is needed.
3.	"MFA prevents every account compromise."	MFA provides extra security, but it cannot stop every attack. In this case, the student approved an unexpected MFA request, which may have allowed access to the account.
4.	"Delete the suspicious logs."	Deleting logs is a bad idea because they are important for investigation. The IT team should save the login records and other details for future checking.
5.	"Send the student's password to support so they can test it."	Passwords should never be shared with anyone, even with technical support. The student should change the password and the IT team should use safe methods to check the account.

D2. Write a better prompt asking an AI tool to analyse only the synthetic evidence, separate facts from assumptions, avoid requesting secrets and state uncertainty. Do not include real student data. (5 marks)

Analyse the given fictional cybersecurity information and help identify possible security risks. Follow these instructions:

1. Use only the information provided in the sample email and synthetic login records.
2. Separate the facts from assumptions.
3. Identify any suspicious login activity and possible security risks.
4. Do not assume that an IP address identifies a particular person.
5. Do not ask for passwords, OTPs or any personal information.
6. Explain any uncertain information clearly.

7. Suggest safe steps to protect the account and investigate the incident.
8. Give simple and clear explanations without making unsupported conclusions.

D3. Write your own corrected 100-150 word security advice. Optional: compare it with an AI tool's output and record what you checked. Tool use is not required for full marks. (6 marks)

Security is very important when using online accounts. If we receive a suspicious email or unexpected login notification, we should not trust it immediately. We should always check the information through official websites or contact the college directly. Passwords and OTPs should never be shared with anyone. MFA gives extra protection, but we should never approve login requests that we did not make. If any unusual login activity is noticed, we should change the password, log out from unknown sessions and inform the IT department. The IT team should keep all login records safely and investigate the issue before making any conclusions. We should also remember that AI answers can contain mistakes, so important security information must always be checked.

D4. Add a short AI-use declaration (tool/purpose or "not used") and two evidence/reference notes supporting your reasoning. (4 marks)

1. AI-Use Declaration:

I used ChatGPT to understand the cybersecurity concepts and organise some of my answers. I also checked the suggestions and made changes according to the given assessment. The final answers are reviewed by me.

2. Evidence and Reference Notes:

Reference 1: CISA – Secure Our World

This reference provides basic information about creating strong passwords, using MFA and identifying phishing messages. It supports my suggestions about keeping accounts safe.

Website: https://www.cisa.gov/secure-our-world

Reference 2: NIST – Generative AI Profile (AI 600-1)

This reference explains that AI-generated information can sometimes be incorrect or misleading. It supports the importance of checking AI answers instead of trusting them blindly